

**Homework Classical Physics, Fall 2023**

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**Problem Set 04.1 (L07 & L08)**

Due on Thursday, 5. October 2023, at 09:30 (in the morning!)

( = Deadline for submitting a pdf file with solution)

**Mechanics - Centrifugal force ( 4 points )**

A sphere hanging on a string of length  $L = 0.50 \text{ m}$  moves with constant uniform motion on a horizontal circular trajectory. The string has an angle of  $32^\circ$  towards the vertical.

- a) What is the velocity of the sphere?
- b) What is the period of the circular motion?